

Project 5: Basic Natural Language Processing with Unsupervised Learning

1. Executive Summary

This project applies basic Natural Language Processing (NLP) and unsupervised learning to the 20 Newsgroups dataset, a collection of online discussion posts from 20 topic groups. The text was cleaned, turned into numbers with Bag of Words, TF-IDF and pre-trained GloVe word embeddings, and grouped with K-Means and hierarchical clustering **without using the topic labels**. The labels were only used afterwards to check how well the clusters match real topics. The trained model was then applied to new, unseen documents.

Question	Answer
How much data was used?	10,992 training posts (after removing 322 empty ones) and 7,296 unseen test posts, across 20 categories
How was text turned into numbers?	Bag of Words and TF-IDF (5,000 words, 99.15% zeros) and 50-dimension GloVe embeddings
How many clusters?	k = 20 : no sharp elbow, silhouette gains slow down after 20, and it matches the 20 newsgroups
Did clear topics appear?	Yes. Hockey, baseball, space, encryption, religion, Middle East, for-sale, PC hardware and more have their own clusters
How close are clusters to the real topics?	NMI 0.317 (K-Means + TF-IDF), 0.314 (hierarchical), 0.333 (K-Means + GloVe, best)
What is the main weakness?	Three general clusters hold 51% of posts; cluster 6 alone holds 27.7%
Does the model work on new text?	Yes. All 5 hand-written examples went to the correct topic, and cluster sizes on the test set match training
How strong is the fit?	Weak. Silhouette 0.0087 (train) and 0.0046 (test); 94.4% of test posts have a fit score below 0.05

Main lesson: unsupervised clustering can find real, readable topics in text without any labels, but the boundaries between topics are soft. Posts with specific vocabulary (encryption, hockey, space) cluster well, while short or conversational posts with common words end up in large "general" clusters.

2. Objective and Data

2.1 Objective

Prepare raw text for analysis, compare different ways of representing text as numbers, apply clustering algorithms to find groups of similar documents, interpret the themes of each group, and test whether the clusters can be used to organize new, unseen documents.

2.2 Dataset

	Training set	Test set
Source	fetch_20newsgroups(subset="train")	fetch_20newsgroups(subset="test")
Documents loaded	11,314	Official test split
Documents after cleaning	10,992	7,296
Categories	20	20
Removed parts	Headers, footers and quoted replies	Headers, footers and quoted replies
Used for	Building features and training clusters	Checking the model on unseen posts

The 20 categories cover computers (5 groups), recreation (autos, motorcycles, baseball, hockey), science (cryptography, electronics, medicine, space), religion (3 groups), politics (guns, Middle East, general) and a for-sale group. Headers, footers and quotes were removed so the model cannot "cheat" by reading the group name or the email address of the sender.

Tools: Python 3, pandas, NumPy, scikit-learn, NLTK, gensim, SciPy, matplotlib, seaborn and Google Colab.

3. Methodology

3.1 Text Preprocessing

- Converted all text to lowercase.
- Removed punctuation and numbers and collapsed extra spaces.
- Split each post into words (tokenization).
- Removed English stop words (common words like "the" and "and") and words shorter than 3 letters.
- Dropped posts that became empty after cleaning (322 in training).

3.2 Text Representation

- **Bag of Words:** counts how many times each word appears in a post.
- **TF-IDF:** weights each word by how often it appears in a post and lowers the weight of words that appear in many posts.
- Both kept the **5,000** most frequent words that appear in at least 5 posts and in no more than 90% of posts.
- **GloVe embeddings:** a pre-trained model (glove-wiki-gigaword-50) gives each word 50 numbers that capture meaning. Each post is represented by the average of its word vectors.

3.3 Clustering

- **Choosing k:** K-Means was run for k = 5, 10, 15, 20, 25 and 30, and inertia and silhouette score were compared.
- **K-Means (k = 20)** on the full TF-IDF matrix, with 10 restarts and random_state = 42.

- **Hierarchical clustering** (Ward linkage) on a 2,000-post sample. The data was first reduced to 100 dimensions with TruncatedSVD (LSA), because Ward linkage needs dense data. The tree was cut into 20 clusters.
- **K-Means on GloVe** document vectors as a bonus comparison.

3.4 Interpretation and Evaluation

- Top terms per cluster, from the K-Means centroids or the average TF-IDF of each hierarchical cluster.
- Dominant true category per cluster, a heatmap of category vs cluster, and a 2D SVD scatter plot.
- **NMI** and **ARI** compare clusters to the true labels (0 = no match, 1 = perfect).
- **Silhouette**, **Davies-Bouldin** and **Calinski-Harabasz** measure cluster quality without labels.
- A custom **fit score** for each post: (separation – cohesion) ÷ the larger of the two, where cohesion is the distance to its own centroid and separation is the distance to the next closest centroid.

4. Key Decisions

Decision	Reason
Remove headers, footers and quotes	Prevents the model from using sender names or group names instead of the actual content
Remove numbers and words under 3 letters	They add noise and very little topic meaning
Limit to 5,000 words (min_df = 5, max_df = 0.9)	Removes rare typos and words found almost everywhere; keeps the matrix manageable
Use TF-IDF for the main clustering	It lowers the weight of common words, so topic words stand out
Choose k = 20	No clear elbow; silhouette gains slow after 20; matches the 20 real newsgroups
Hierarchical on a 2,000-post sample	Ward linkage on all 10,992 posts is too slow and memory-heavy
Reduce to 100 dimensions before Ward	Ward needs dense data; LSA keeps the main topic patterns
Use labels only for evaluation	Keeps the method truly unsupervised
Test on the official test split	Shows how the model behaves on posts it has never seen

5. Analysis and Results

5.1 Text Preparation

Preprocessing turned each post into a clean list of words. For example, the first post about a car ("I was wondering if anyone out there could enlighten me on this car...") became *wondering*, *enlighten*, *car*, *saw*, *day*, *door*, *sports*, *car*, *looked*, *late*, *early*, *called*. The most common words across the corpus were general words such as *people* (3,997), *like* (3,885), *don't* (3,862), *just* and *know*.

One strange word, **maxaxaxaxaxaxaxaxaxaxaxaxaxax**, appears 3,317 times. It is not a real word: it comes from encoded files that were pasted into posts. It is a good example of noise that basic cleaning does not catch.

5.2 Text Representation

A small example with three sentences shows the difference between the two methods. Bag of Words simply counts words. TF-IDF gives "the" a lower weight (0.32) than words unique to one sentence like "fast" (0.53), because "the" appears in every sentence.

Method	Features	Type	Captures meaning?
Bag of Words	5,000	Sparse counts	No
TF-IDF	5,000	Sparse weights	No
GloVe average	50	Dense vectors	Yes (similar words are close)

The full matrices have 10,992 rows and 5,000 columns, and **99.15%** of the values are zero, because each post uses only a small part of the vocabulary. TF-IDF already shows clear topic words per category, for example *key*, *encryption*, *clipper* for sci.crypt and *space*, *nasa*, *orbit* for sci.space.

GloVe embeddings capture meaning. The words most similar to *car* are *truck*, *cars*, *vehicle*; *car* vs *automobile* has a similarity of 0.696, while *car* vs *banana* is only 0.219; and *king* – *man* + *woman* gives *queen*. One limitation shows too: the closest words to *windows* are *window*, *console*, *doors*, because GloVe was trained on general news and does not know that in this dataset Windows usually means the operating system.

5.3 Choosing the Number of Clusters

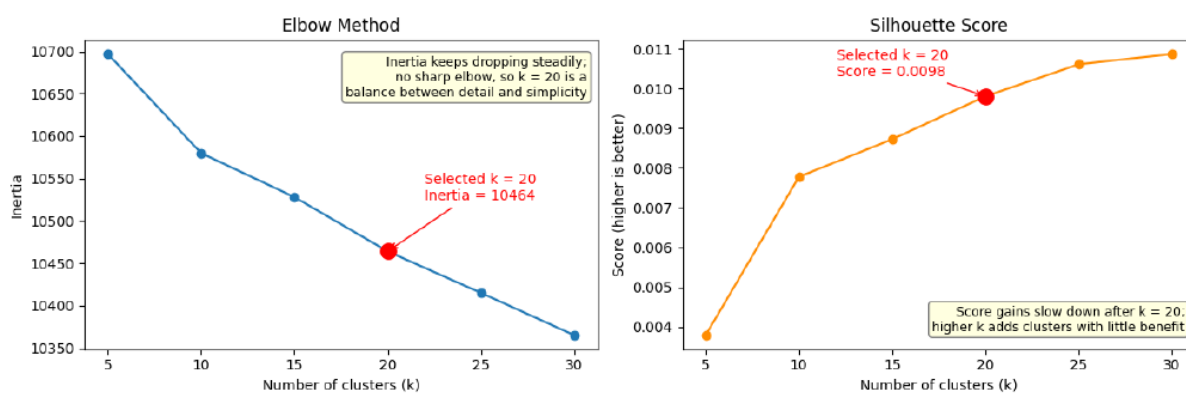


Figure 1. Elbow method (inertia) and silhouette score for k = 5 to 30.

k	5	10	15	20	25	30
Inertia	10,697	10,580	10,528	10,464	10,415	10,365
Silhouette	0.0038	0.0078	0.0087	0.0098	0.0106	0.0109

Inertia drops steadily with no clear "elbow", which is common with text data. The silhouette score rises quickly up to $k = 10$ and then grows more slowly. $k = 20$ was chosen as a balance between detail and simplicity, and because it matches the 20 real newsgroups, which makes comparison easier. All silhouette scores are very low (about 0.01): in 5,000 dimensions, posts are almost the same distance from every centroid, so this is expected for sparse text.

5.4 K-Means Clusters and Their Themes

Cluster	Size	Top terms	Theme
0	71	gebcadredslpittedu, chastity, jxp, shameful, intellect	Repeated email signature (sci.med)
1	207	space, launch, nasa, lunar, orbit, moon	Space
2	1,227	use, using, software, need, program, data	General computing
3	168	israel, israeli, jews, arab, arabs	Middle East politics
4	398	car, bike, cars, engine, miles	Cars and motorcycles
5	421	god, jesus, bible, faith, believe	Christianity
6	3,042	new, like, did, article, time, good	General / short posts
7	253	card, video, drivers, monitor, vga	Graphics cards and drivers
8	392	windows, file, files, dos, program	MS Windows and DOS
9	250	year, hit, braves, pitching, season	Baseball
10	524	thanks, email, advance, know, looking	Requests for help
11	282	sale, offer, shipping, condition, price	Items for sale
12	788	people, government, think, gun, right	Politics and guns
13	1,339	dont, just, think, like, know, really	General opinion
14	438	game, team, games, players, hockey, nhl	Hockey
15	150	window, application, manager, motif, xterm	X Window System
16	268	key, encryption, chip, clipper, escrow	Encryption
17	262	drive, scsi, drives, hard, ide, disk	Hard drives
18	110	armenian, turkish, armenia, genocide	Armenia and Turkey
19	402	does, know, simms, like, anybody, help	Hardware questions

Most clusters have a clear, readable theme, and several match one newsgroup almost exactly: hockey (14), space (1), encryption (16), Christianity (5), for-sale (11) and Middle East (3 and 18). The weak point is the three general clusters **6, 13 and 2**, which together hold **5,608 posts (51%)**. Their top words (*new, like, dont, just, use*) are general conversation words, not topic words.

Cluster 0 is a special case. Its top words come from one person's email signature that was repeated in many posts. The removal of footers did not catch it, so K-Means created a whole cluster for it.

5.5 Comparing Clusters with the Real Categories

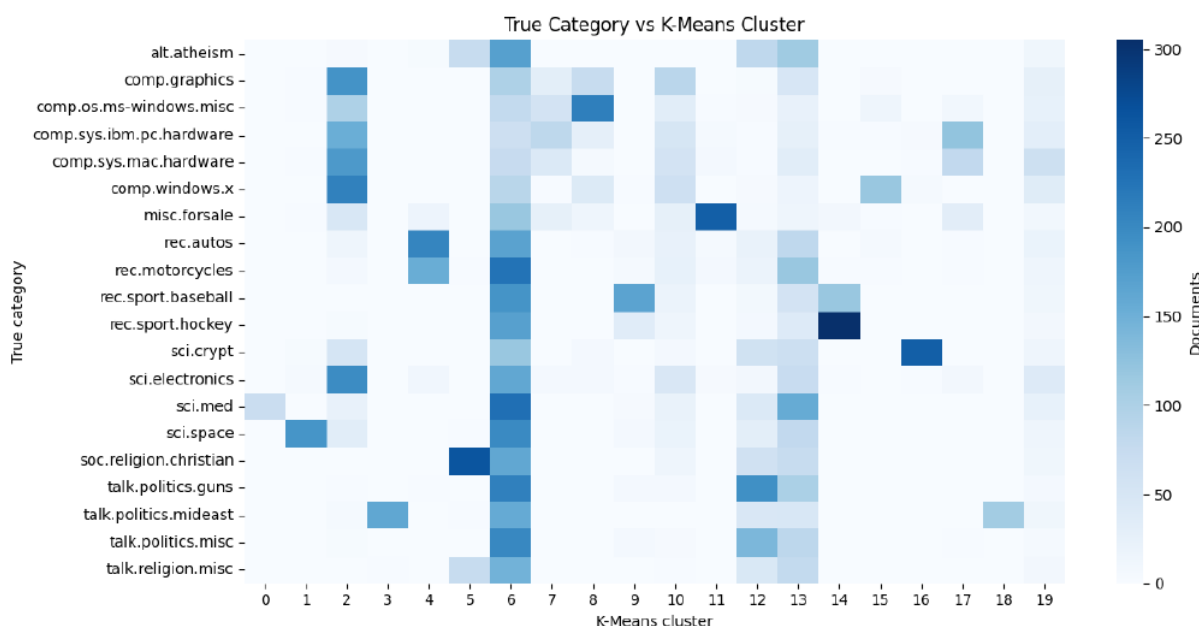


Figure 2. Number of posts from each true category in each K-Means cluster.

A dark cell means many posts from that category went to that cluster. Strong single cells confirm the clean topics: rec.sport.hockey → 14, soc.religion.christian → 5, misc.forsale → 11, sci.crypt → 16 and sci.space → 1. The five computer groups share clusters 2, 7, 8, 15, 17 and 19, because they use very similar vocabulary. The **column for cluster 6 is dark for almost every category**, which shows it is a "catch-all" cluster rather than a topic.

Metric	Value	Meaning
Adjusted Rand Index (ARI)	0.080	Low: many pairs of posts from the same category end up in different clusters, mostly because of the large general clusters
Normalized Mutual Information (NMI)	0.317	Moderate: the clusters carry real information about the topic

5.6 Visualizing the Clusters in 2D

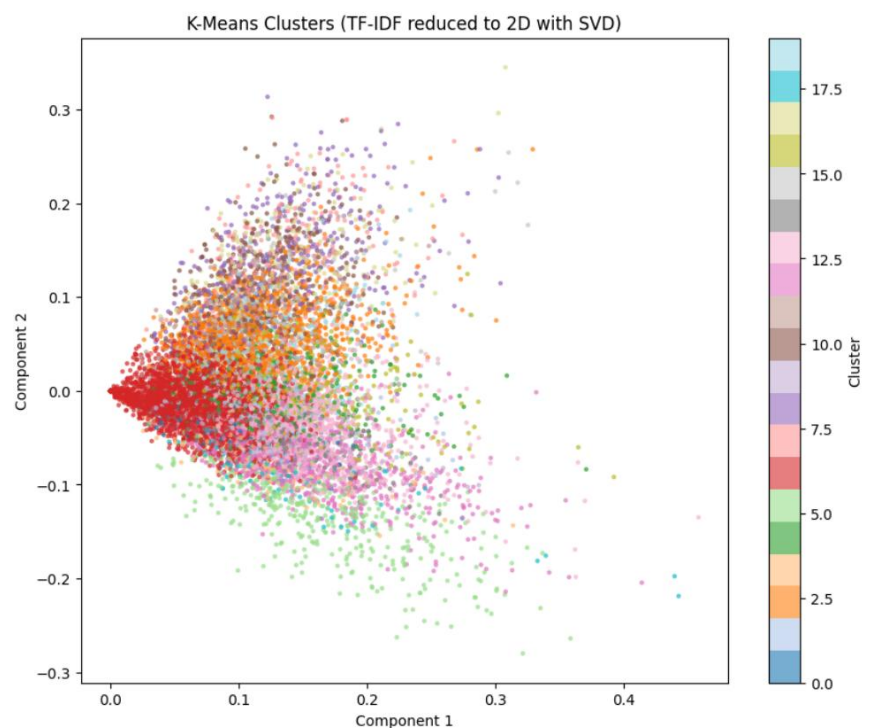


Figure 3. K-Means clusters after reducing TF-IDF to 2 dimensions with SVD.

The clusters overlap strongly in 2D. The dense red area near (0, 0) is mostly the general cluster 6: short posts with few distinctive words sit close to the center of the space. Topic clusters spread outwards in different directions. Because only 2 of 5,000 dimensions are shown, the plot understates how separate the topic clusters really are, but it clearly shows that there are no sharp borders between groups.

5.7 Hierarchical Clustering

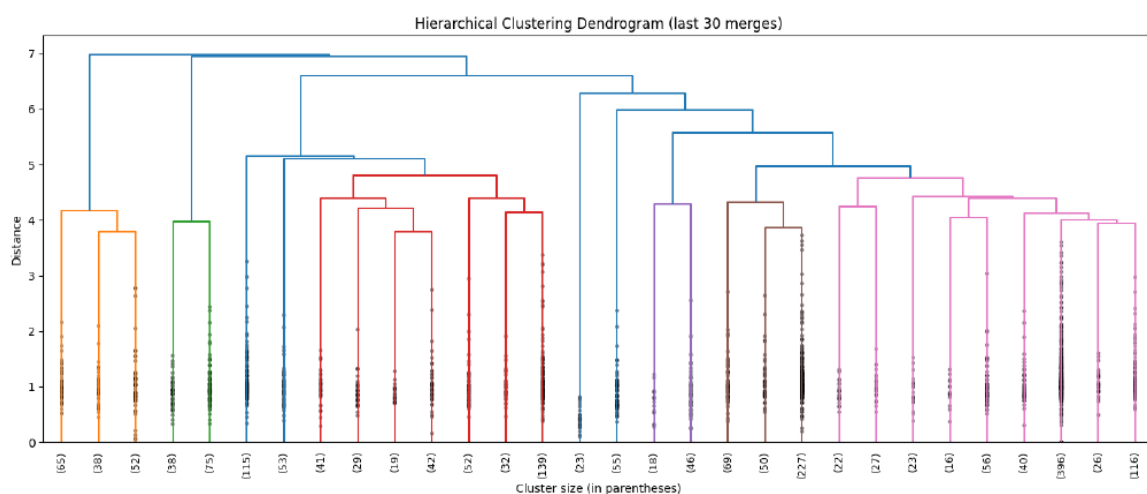


Figure 4. Ward dendrogram of the 2,000-post sample (last 30 merges).

The dendrogram shows how groups merge as the distance grows. Small, tight groups (for example the orange and green branches on the left) join late, which means they are clearly different from the rest. The large pink branch on the right contains many groups that merge at similar heights, which matches the general posts seen with K-Means.

Cluster	Size	Top terms
1	155	game, team, year, season, play, games, fans
2	113	god, church, jesus, bible, christian, faith
4	53	drive, scsi, tape, controller, drives, floppy
6	29	window, motif, server, xterm, pixmap
11	55	chip, key, clipper, keys, encryption, nsa
12	18	turkey, armenian, turkish, armenians, genocide
13	46	israel, jewish, israeli, jews, peace, war
17	27	condition, miles, sale, excellent, asking, shipping
18	23	space, lunar, orbit, mission, launch, moon
20	578	use, know, just, new, email, like, dont

Hierarchical clustering found almost the same themes as K-Means, including the signature cluster (cluster 10). One difference: it merged hockey and baseball into one sports cluster (1), while K-Means kept them apart. It also has one large general cluster (20) with **578 posts (28.9%)**. Its NMI of **0.314** on the sample is almost the same as K-Means (0.317), so both methods find the same structure.

5.8 Comparison of Methods

Method	Posts	NMI vs true labels
K-Means + TF-IDF	10,992	0.317
Hierarchical (Ward) + LSA	2,000	0.314
K-Means + GloVe	10,992	0.333

K-Means on GloVe embeddings scored best, even though it uses only 50 numbers per post instead of 5,000. Embeddings group words with similar meaning (for example *car* and *vehicle*), so two posts can be close even if they use different words. The gain is small, but it shows that meaning-based features help.

5.9 Predicting Clusters for New Documents

New document	Cluster	Closest category	Distance	Next closest
The goalie made an amazing save...	14	rec.sport.hockey	0.962	1.001
NASA launched a new satellite...	1	sci.space	0.970	0.998
My graphics card driver keeps crashing...	7	comp.sys.ibm.pc.hardware	0.920	0.968
The government should not restrict... firearms	12	talk.politics.guns	0.995	1.000

New document	Cluster	Closest category	Distance	Next closest
Encryption keys and the clipper chip...	16	sci.crypt	0.919	1.001

New text is cleaned with the same preprocessing function, turned into TF-IDF with the already trained vectorizer, and assigned to the nearest centroid. All five examples went to the correct topic. The gap between the nearest and next closest centroid shows confidence: the encryption post has a clear gap (0.919 vs 1.001), while the firearms post is almost tied (0.995 vs 1.000), because cluster 12 mixes politics and guns.

5.10 Validation on Unseen Test Documents

Metric	Train	Test (unseen)
Silhouette (higher better)	0.0087	0.0046
Davies-Bouldin (lower better)	8.2536	7.9985
Calinski-Harabasz (higher better)	6.3438	5.3642
Mean cohesion (lower better)	0.9732	0.9780
Mean separation (higher better)	0.9910	0.9909
Mean fit score (higher better)	0.0192	0.0144
NMI vs true labels	0.3174	0.2844
ARI vs true labels	0.0800	0.0632

The test results are slightly weaker than training on most metrics, but in the same range. NMI drops from 0.317 to 0.284, so the model **generalizes reasonably well** and is not just memorizing the training posts. Davies-Bouldin is even slightly better on test. The main issue is not overfitting but weak cluster separation overall: mean cohesion (0.978) and separation (0.991) are almost equal, so a typical post is only slightly closer to its own cluster than to the next one.

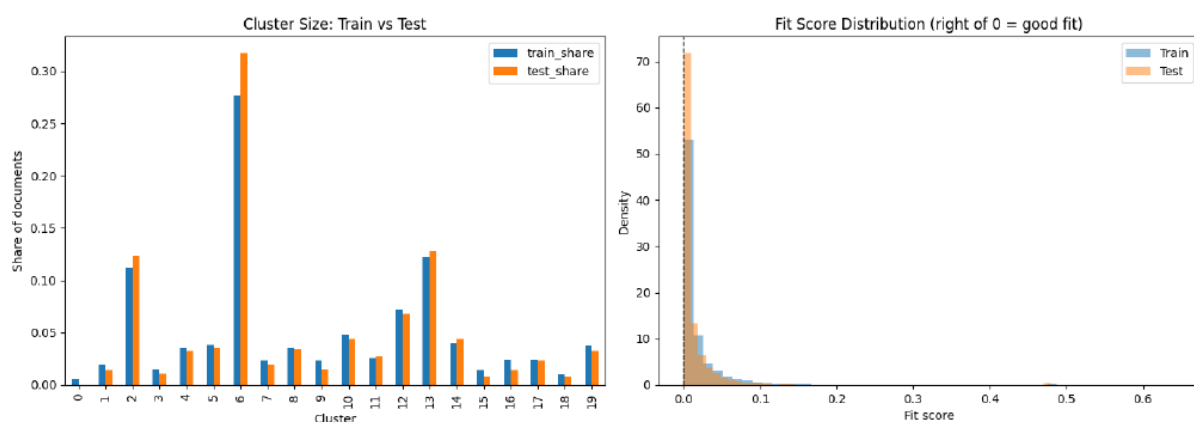


Figure 5. Cluster size on train vs test (left) and fit score distribution (right).

Cluster sizes are stable: each cluster gets a similar share of test posts as training posts. Only the general cluster 6 grows (27.7% → 31.7%), and the signature cluster 0 disappears on test (0.6% → 0.0%), which confirms it was based on one person's signature, not a topic.

Per-cluster fit: the tightest clusters on test are the specific ones: 0 (cohesion 0.863), Armenia/Turkey 18 (0.906), hard drives 17 (0.940) and Middle East 3 (0.946). The loosest are the general clusters 2 (0.988), 6 (0.987) and 13 (0.986).

5.11 Best and Worst Fitting Test Documents

	Cluster	Fit	True category	Text (start)
Best	6	0.480	comp.windows.x	"please subscribe me."
Best	6	0.480	rec.autos	"cheek."
Best	6	0.480	alt.atheism	"It is still incestuous.... :)"
Worst	13	0.000	sci.electronics	"Can someone tell me where to find 120volt 3 watt 40 mA fans..."
Worst	9	0.000	rec.sport.baseball	"With all the recent problems the Indians have been having..."

This result needs care. The five "best" fits are all very short posts in cluster 6 from five different categories. They have almost no words in the vocabulary, so they sit close to the center of the space, exactly where cluster 6's centroid is. A high fit score here means "clearly generic", not "clearly on topic". The worst fits are real, topic-rich posts that sit between two clusters, for example a baseball post that went to cluster 9 but is just as close to another cluster. Overall, **94.4%** of test posts have a fit score below 0.05, which confirms that the cluster borders are soft.

6. Insights

- **Topics can be found without labels.** Hockey, space, encryption, religion, Middle East and for-sale posts form clear clusters with readable top words.
- **Specific vocabulary makes clean clusters.** Groups with unique words (clipper, nhl, nasa) cluster well; groups that share words (the five computer groups, the three religion groups) get mixed.
- **General clusters are the main weakness.** Half of the posts fall into three clusters built on everyday words, which keeps ARI low (0.080).
- **Meaning helps.** GloVe embeddings reach the best NMI (0.333) with 100 times fewer features than TF-IDF.
- **The model generalizes.** Test NMI (0.284) and cluster shares are close to training, and all five hand-written examples were assigned correctly.
- **A high fit score can be misleading.** The "best" fitting posts were one-line messages, so fit scores should be read together with post length and top terms.
- **Cleaning is never finished.** An email signature and encoded file text survived preprocessing and created a cluster and a top word of their own.

7. Limitations

- Silhouette scores are very low (about 0.01), so the clusters are soft and overlap strongly.
- The 20 real categories overlap in content (for example the religion and computer groups), so a perfect match is not possible.
- Hierarchical clustering was only run on a 2,000-post sample, so its NMI is not directly comparable to the full-data K-Means.
- Averaging GloVe vectors ignores word order, and GloVe was trained on general news, not forum text (for example it links "windows" to doors).
- Very short posts carry almost no information but still get a cluster and can get a high fit score.
- $k = 20$ were chosen partly because there are 20 newsgroups; with no labels, a different k could be just as reasonable.

8. Recommendations and Next Steps

- **Clean more noise:** remove repeated signatures, encoded text (such as "maxaxax") and very short posts (for example under 10 words) before clustering.
- **Add lemmatization** so that words like "drive", "drives" and "driving" count as one word.
- **Cluster on LSA-reduced TF-IDF** (for example 100 dimensions) for the full dataset, which usually gives cleaner K-Means clusters than the raw sparse matrix.
- **Try topic models** such as LDA or NMF, which allow a post to belong to more than one topic.
- **Use better embeddings**, such as TF-IDF-weighted GloVe or sentence-transformer models, which capture the meaning of whole sentences.
- **Flag low-confidence predictions** for new documents when the nearest and next closest centroids are almost tied.

9. Conclusion

This project took 10,992 raw newsgroup posts, cleaned them, and represented them with Bag of Words, TF-IDF and GloVe embeddings. K-Means with $k = 20$ and hierarchical clustering both found clear, readable topics such as hockey, space, encryption, religion and the Middle East, reaching an NMI of about 0.31 against the true labels, and 0.333 with GloVe embeddings. The model assigned all five new example documents correctly and behaved similarly on 7,296 unseen test posts, with stable cluster sizes and only a small drop in NMI ($0.317 \rightarrow 0.284$). The main weakness is that half of the posts fall into general clusters built on everyday words, and that cluster borders are soft (94.4% of test posts have a weak fit). The key lesson is that **unsupervised text clustering is useful for discovering and organizing topics, but the results depend strongly on how the text is cleaned and represented.**

Appendix: Notebook and Reproducibility

- **Notebook:** Project.5_Basic_NLP_with_Unsupervised_Learning.ipynb (link in the page header).
- **Data:** 20 Newsgroups, loaded with scikit-learn's fetch_20newsgroups (train and test subsets), with headers, footers and quotes removed.

- **Embeddings:** glove-wiki-gigaword-50, downloaded with gensim.downloader.
- **Settings:** TF-IDF with max_features = 5,000, min_df = 5, max_df = 0.9; K-Means with k = 20, n_init = 10; all random steps use random_state = 42.
- **To reproduce:** open the notebook in Google Colab and run **Runtime → Restart session and run all**.